

Designing a Gamified Task Map Based on Multisensory User-Contributed Data to Promote Urban Exploration

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Abstract

Interactive systems supporting everyday urban movement increasingly aim to emphasize not only efficiency but also exploratory experiences that reflect how users perceive and experience urban spaces. In this context, multisensory place information, such as visual, auditory, and olfactory impressions of urban environments, has been shown to play an important role in shaping subjective urban experiences. However, it remains unclear how such multisensory information can be operationalized within interactive systems to actively promote urban exploration. The aim of this paper is to promote urban exploration through the design of a gamified task map. We present **Spot Hunting**, an exploration-oriented system that transforms user-contributed multisensory place information into goal-driven exploration tasks, referred to as *Quests*. Rather than providing route guidance, the system supports exploratory urban movement through task-based interaction. Spot Hunting addresses this challenge by generating exploration tasks based on multisensory cues associated with specific locations, including visual focus point, sounds, and smells. To demonstrate our design, we generated a task map consisting of 11,855 Quests based on multisensory place information and conducted a field deployment with 31 participants in an urban environment. We describe the design rationale of Spot Hunting and report findings from a multi-month field deployment, analyzing users' movement patterns and interaction behaviors to explore how multisensory-based tasks shape exploratory urban behavior in everyday contexts. The results suggest that presenting multisensory place information as lightweight, gamified exploration tasks, can effectively promote urban exploration while maintaining user autonomy. This work illustrates how subjective urban experience data can be integrated into interactive systems to support exploration-oriented urban movement.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **HCI design and evaluation methods**; • **Information systems** → *Location based services*.

Keywords

Urban exploration, Gamified task map, Multisensory place experience, Interaction design

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1 Introduction

Urban navigation systems have traditionally focused on optimizing efficiency, such as minimizing travel distance or time. However, recent studies suggest that everyday urban movement also plays an important role in shaping users' subjective experiences, including comfort, curiosity, and emotional engagement with their surroundings. As a result, there is growing interest in systems that support urban exploration and encourage users to discover new places and engage more deeply with urban environments.

In this context, multisensory place experiences, such as visual, auditory, and olfactory impressions, have been shown to play an important role in shaping how people perceive and experience urban environments [1, 3, 5, 6]. These multisensory experiences have also been linked to psychological outcomes, such as attention restoration, stress reduction, and emotional well-being. However, much of the existing research has primarily focused on analyzing or visualizing multisensory information, with limited attention to how such subjective experiences can be operationalized within interaction design to support users' movement behaviors during everyday urban activities.



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Figure 1: Relationship between QuestPOST and Spot Hunting. Multisensory place experience data collected via QuestPOST (prior work [13]) are aggregated and transformed into exploration Quests in Spot Hunting, the exploration support system proposed in this paper.

The aim of this research is to promote urban exploration by incorporating multisensory place experiences into the design of an interactive system that supports exploratory urban movement. Recent large-scale field studies have demonstrated the feasibility of collecting multisensory place impressions in urban environments and analyzing their relationship to urban context and human perception [9, 10, 13]. While these studies provide a foundation for understanding multisensory urban experiences, experiential data alone are insufficient to actively encourage exploratory movement. To promote urban exploration, a key design challenge lies in how exploration tasks are defined within urban spaces. Prior research has further suggested that users value opportunities to explore unfamiliar environments without being constrained by shortest-path guidance [8]. In addition, recent work has shown that task-based and gamified interactions can effectively encourage spatial exploration by reframing familiar environments as sites for discovery [11, 12].

To address this challenge, we present **Spot Hunting**, an interactive system featuring a gamified task map that transforms multisensory place experiences into goal-driven exploration tasks. By defining exploration tasks based on multisensory cues associated with specific locations, rather than providing route guidance or prescribing optimal routes, Spot Hunting supports both voluntary exploration and task-based interaction. This distinguishes Spot Hunting from generic point-based map overlays by grounding tasks in subjective place experiences.

2 From Multisensory Data to Exploratory Urban Movement

This section explains how multisensory place experience data are collected in real-world urban environments and how such data are transformed into interactive mechanisms that support exploratory urban movement. Our approach consists of two stages. First, multisensory place impressions are collected through QuestPOST, a mobile-based data collection system developed in our prior work. Second, the collected data are utilized in Spot Hunting to generate gamified exploration tasks that encourage users to engage in exploratory movement beyond habitual routes.

Figure 1 provides an overview of the relationship between QuestPOST and Spot Hunting. QuestPOST serves as a data collection platform that gathers user-contributed multisensory place experiences, while Spot Hunting builds on the aggregated data to support exploratory urban movement through task-based interaction. In

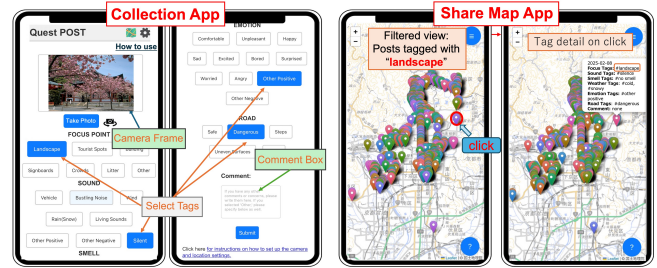


Figure 2: User interaction flow in our QuestPOST system consists of the posting and visualization components. (Left) Photo and tag submission using the Collection App. (Right) Tag-based visualization and detailed view in the Share Map App.

this process, the aggregated multisensory data are transformed into exploration **Quests**, which function as the basic task units in Spot Hunting. Each Quest is generated from previously submitted place posts that include geo-tagged photographs and associated multisensory tags. A Quest inherits the original photo and selected sensory cues as exploration hints, and defines a coarse-grained target area around the original posting location.

2.1 QuestPOST: Multisensory Place Experience Collection

QuestPOST is a mobile-based data collection system designed to gather multisensory place experience data in real-world urban environments. Through everyday use during walking or commuting, users contribute geo-tagged posts by capturing photos at visited locations and annotating them with subjective sensory impressions.

As shown in Figure 2, QuestPOST supports the annotation of multiple sensory dimensions associated with a place. Users select tags related to visual focus points (e.g., landscape, buildings), sounds (e.g., vehicles, living sounds), smells, emotions, weather, and road conditions, and may optionally add free-text comments. Each submission therefore captures not only the physical appearance of a location but also how it is subjectively perceived and experienced in situ.

All submitted posts are stored together with their location information and aggregated as multisensory place experience data.

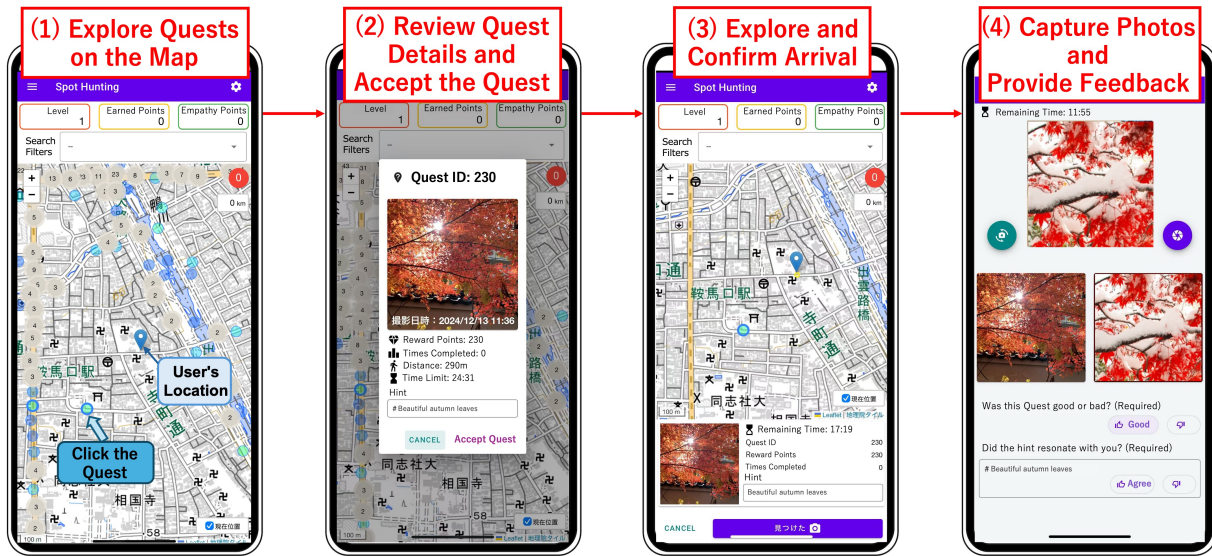


Figure 3: Operation flow of the Spot Hunting system for task-based urban exploration using Quests generated from multisensory place experience data.

Posts meeting basic quality criteria, such as being captured in outdoor environments, containing complete sensory tags, valid geo-information, and excluding identifiable persons, are retained and used as candidate locations for generating exploration Quests.

2.2 Spot Hunting: Gamified Task Map for Urban Exploration

Spot Hunting is an exploration-oriented guidance system that builds on multisensory place experience data to encourage users to engage in voluntary and reflective urban exploration at the level of goal and task selection. Rather than optimizing for efficiency or providing explicit route guidance, the system is designed to subtly reshape everyday movement by introducing uncertainty, spatial constraints, and lightweight task-based incentives.

Figure 3 illustrates the interaction flow of Spot Hunting from Quest discovery to completion and reflection. First, users browse available exploration Quests on a map (1). Quests are displayed as circular markers with ambiguous target areas, allowing users to explore potential destinations without revealing exact locations. The map also indicates the user's current position, enabling users to discover Quests in surrounding areas while encouraging movement beyond their immediate vicinity. Next, by selecting a Quest marker, users can review detailed information about the Quest (2). This includes a reference photograph, multisensory hints derived from prior place experiences, and basic task information such as coarse-grained estimated distance and reward points. At this stage, users may freely decide whether to accept the Quest or continue browsing without obligation. After accepting a Quest, users begin exploration toward the target area (3). Rather than providing turn-by-turn navigation or precise coordinates, the system supports exploration through a map view that preserves spatial ambiguity. Users rely on environmental cues and their interpretation of the reference image and hints to locate the place, and can confirm arrival

once they believe they have reached the target area. Finally, upon Quest completion, users capture a photograph of the location and provide reflective feedback on their experience (4). This includes a simple evaluation of the Quest, assessment of resonance with the original multisensory hints, and optional comments. Through this process, Spot Hunting integrates physical exploration with reflective interaction, linking prior place experiences with newly generated ones.

Quest Representation and Discovery. Exploration targets in Spot Hunting are represented as Quests, each automatically generated from a previously collected multisensory place experience. Quest generation is rule-based: each qualifying geo-tagged post is transformed into a Quest. The original photograph serves as a visual cue, while selected sensory tags are presented as concise multisensory hints.

Quests are displayed on a map as circular markers with intentionally ambiguous target areas (approximately 20 meters in radius). Exact coordinates and route guidance are not provided, requiring users to rely on visual cues and environmental interpretation during exploration. To further encourage physical movement, Quests within a 100-meter radius of the user's current location are hidden, preventing immediate task completion and promoting exploration beyond habitual surroundings.

Quest Acceptance and Non-linear Exploration. Users may freely browse, accept, cancel, or ignore Quests without penalty, and multiple Quests can be pursued simultaneously. Quests are persistent and shared: while each user may complete a Quest only once, the same Quest can be completed by multiple users. This design emphasizes autonomy and supports non-linear exploration, framing Quests as shared urban exploration resources rather than consumable objectives.

Quest Completion and Reflective Feedback. A Quest is completed only after the user physically reaches the target area and captures a new photograph of the location. This requirement serves as an embodied verification step, ensuring that task completion is grounded in real-world interaction rather than proximity alone. Upon completion, users are asked to reflect on their experience by providing multiple forms of feedback. Specifically, users evaluate whether the Quest was perceived as “good” or “bad,” indicate their level of resonance with the original multisensory hints associated with the Quest, and optionally provide a short textual comment. In addition, users complete several five-point Likert-scale evaluations covering multisensory impressions (e.g., smell, sound, weather), emotional response, perceived safety, route comfort, route stimulation, and perceived task difficulty. These evaluations capture how the place was experienced during exploration and how the journey itself contributed to the overall impression.

Lightweight Gamification and Motivation. Completing Quests awards points that contribute to the user’s level within the system. This gamification is intentionally lightweight and non-competitive, serving as gentle motivation rather than performance pressure. By avoiding rankings or leaderboards, Spot Hunting prioritizes curiosity-driven exploration over optimization.

Design Implications for Exploratory Urban Movement. By combining task-based interaction, spatial ambiguity, optional engagement, and reflective feedback, Spot Hunting operationalizes multisensory place experience data as a mechanism for supporting exploration-oriented urban movement. The system encourages users to deviate from habitual routes and engage more attentively with their surroundings while maintaining full control over when and how they explore.

3 Exploration-Promoting Effects of Task-Based Design

This section reports observed effects of Spot Hunting on users’ exploratory behavior in real-world urban environments. Rather than evaluating individual interface components in isolation, we focus on how task-based interaction and related design elements collectively shaped exploration behavior, based on system logs and Quest completion records.

A total of 31 participants took part in a real-world field deployment of Spot Hunting conducted in an urban environment. The deployment took place over a multi-month period, from late September 2025 to mid-January 2026, allowing participants to engage with the system during everyday walking and commuting over time. Based on multisensory place experience data collected through prior deployments, the system generated 11,855 exploration Quests across the target area. During the deployment period, participants completed 1,591 Quests in total. Quest completions occurred across multiple days and spatial locations, indicating repeated engagement with the system rather than one-time interaction. Completed Quests were spatially distributed across a wide urban area and spanned multiple neighborhoods, rather than being concentrated in a single location.

3.1 Observed Exploration Behavior

Analysis of Quest acceptance and completion logs suggests that participants frequently engaged in exploration beyond immediately nearby or convenience-based movement. Specifically, completed Quests were spatially distributed across the target area rather than concentrated around users’ initial locations. In many cases, participants accepted and completed multiple Quests within a single walking session, indicating sustained engagement with exploration-oriented behavior.

Furthermore, Quest interactions were often interwoven with everyday movement, such as commuting or routine walking. Rather than initiating exploration as a separate activity, users incorporated Quests into ongoing mobility, suggesting that the task-based design supported exploration without requiring explicit planning or route optimization.

3.2 Contribution of Key Design Elements

Observed user behavior suggests that several design elements contributed to exploration promotion.

First, the task-based structure itself played a central role. By framing multisensory place experiences as optional Quests rather than predefined destinations or routes, the system encouraged users to reinterpret familiar urban spaces as potential exploration targets. This effect was reflected in users’ willingness to engage with Quests during routine movement rather than only in unfamiliar areas.

Second, spatial ambiguity was closely associated with observed indirect movement patterns. The use of imprecise target areas and the absence of turn-by-turn navigation required users to rely on environmental cues and personal interpretation. Interaction logs and completion patterns indicate that users often approached target areas via indirect paths, suggesting route deviation and localized exploration rather than direct navigation.

In addition, the non-linear and voluntary nature of Quest interaction supported exploration by reducing pressure to optimize for efficiency. Users could accept, cancel, or ignore Quests without penalty, and multiple Quests could be pursued in parallel. In contrast, lightweight gamification elements such as points and levels primarily functioned as secondary motivation. While they supported continued engagement, observed behavior suggests that exploration was driven more strongly by task content and spatial discovery than by reward accumulation.

3.3 Summary

Overall, the results suggest that transforming multisensory place experience data into a task-based exploration mechanism can effectively support urban exploration in everyday contexts. Rather than directing users along optimal routes, Spot Hunting promoted exploration through optional, spatially ambiguous tasks that users could integrate into ongoing movement. Among the system’s design elements, the task-based representation of exploration opportunities and spatial ambiguity were most closely associated with exploratory behavior observed during the deployment.

4 Conclusion

This paper presented Spot Hunting, an exploration-oriented guidance system that transforms multisensory place experience data

into gamified exploration tasks. Rather than guiding users along optimal routes, Spot Hunting encourages voluntary and reflective urban exploration by introducing task-based interaction, spatial ambiguity, and lightweight incentives grounded in subjective place experiences.

Through a real-world field deployment, we observed that presenting exploration opportunities as optional tasks was associated with exploratory movement beyond habitual or efficiency-driven routes. In particular, the task-based structure itself and the deliberate use of spatial ambiguity played a central role in encouraging users to deviate from familiar paths, engage more attentively with their surroundings, and integrate exploration into everyday urban mobility. While lightweight gamification supported continued engagement, exploration behavior appeared to be driven primarily by curiosity toward task content and spatial discovery rather than by competitive rewards.

Overall, our findings suggest that task-oriented interaction grounded in multisensory place experiences can serve as an effective approach for promoting urban exploration without imposing prescriptive navigation guidance. Spot Hunting demonstrates how experiential urban data can be operationalized as an interactive mechanism to support exploration-oriented urban movement, offering an alternative perspective to efficiency-focused urban mobility systems.

While this study focused on log-based behavioral analysis, future work will incorporate subjective questionnaire measures to deepen understanding of users' experiential outcomes. In particular, we plan to analyze survey data including affective states (e.g., Positive Affect and Negative Affect, measured using PANAS [4]), psychological well-being (PERMA-Profilers [2]), and sensation-seeking tendencies (Brief Sensation Seeking Scale; BSSS [7]) to examine how individual traits and emotional responses relate to exploration behavior. In addition, we aim to develop predictive models that integrate multisensory cues and user characteristics to better understand and support personalized exploration-oriented urban movement.

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